

Roshan Kumar Gupta

Research Engineer & Systems Architect

Chandigarh, India

+91 9334098565

✉ roshankumargupta.sh@gmail.com

🌐 roshankumargupta.tech

Education

2025–2029 **Bachelor of Technology in Computer Science**, Chandigarh University, Mohali, Expected
Relevant Coursework: High-Performance Computing, Low-Level Systems, Parallel Architectures, Database Systems.

Experience

- Jul 2026–Present **Author & Lead Architect**, Earthos Platform, earthos.shop
- Authored *The AGI Question: What Does It Take to Become a Mind?* — exploring cognitive philosophy, neuroscience, and computational intelligence.
 - Engineered production storefront and reader workspace (earthos.shop) using Next.js 16 and Supabase PostgreSQL.
 - Developed automated nearest-node multi-warehouse logistics and secure real-time DRM watermark fingerprinting.
- Dec 2025–Jan 2026 **L3 IT Intern**, SMG Electric Scooters Ltd, Chandigarh
- Designed and developed a custom internal CRM portal, streamlining logistics and vendor development workflows for enterprise-scale operations.
 - Collaborated with cross-functional teams to identify infrastructure bottlenecks, reducing manual processing time through automation.
- Oct 2025–Sep 2026 **Student Representative**, Chandigarh University, Mohali
- Acted as primary student-admin liaison, managing high-stakes communication to resolve complex infrastructure and academic issues.
 - Facilitated multiple departmental town-halls, ensuring transparent information flow and collaborative problem-solving across 400+ student cohorts.

System Engineering

- Project Earthos **Project Earthos (Synapse Arch)** | *Autonomous Cognitive Substrate*
- Architected and implemented Synapse Arch, a neural-symbolic cognitive substrate combining event-sourced memory hydration and global workspace attention (GWT) coalitions.
 - Implemented AST and import locks to block non-deterministic floating-point namespaces, enforcing strict byte-level execution parity across distinct CPU platforms.
- FieldChain **FieldChain** | github.com/rk-roshan-kr/FieldChain
- Secured a \$1,500 research grant; engineered a GPU-centric storage integrity layer sustaining 7GB/s sequential writes via PCIe 4.0, outperforming specialized DPUs by 4.2x (166.86 GiB/s peak bandwidth) via Vulkan.
 - Awarded **Best Student Paper** at IEEE CCNCPS 2026 (Dubai, UAE) for GPU-accelerated storage integrity research.
- Project TARS **Project TARS Ecosystem** | tars.earth
- Secured \$5,500 Emergent Ventures grant; designed the TARS Core physics-constrained validation framework for sparse two-transit exoplanet candidate vetting, achieving 70.1% precision on NASA TESS data.
 - Formulated architectural specifications for the TARS-EX (physics-aware ensemble ML) research branch.
 - Engineered *tars_infra*, a parallel FITS data ingestion engine utilizing multi-drive spanning to download and index 150,000+ files from the MAST database.
- QUANTA **Thesis QUANTA** | Working Paper
- Defined an operational info-horizon for NISQ circuits via entropic noise metrics; formulated exponential scaling law and mapped the "Undead Zone" to balance state extractability against environmental decoherence.
- HSA* / Arch **HSA* Algorithm & Roshan Arch**
- Formulated routing engine using inverse square law safety weights, achieving pathfinding in 97μs (140x faster than Dijkstra).
 - Architected a custom deterministic memory framework in C++ to bypass garbage collection pauses and ensure predictable real-time runtimes.
- Cryptic **Cryptic** | ZeroToOne Hackathon Winner
- Served as Frontend Lead for a DeFi Treasury dashboard, managing UI/UX architecture and 3D WebGL spatial liquidity flow visualizations.

Technical Skills

Low-Level Systems CUDA, Vulkan, PTX, C++, Deterministic Runtimes, Memory Architectures.
React (Architecture), PyTorch (Physics-ML), Figma (UI Primitives).

Honors & Certifications

Awards **Best Student Paper Award** (IEEE CCNCPS 2026, Dubai); **Emergent Ventures Grant** (\$5,500); **1st Place**, Tekathon 2026 / SIH Internal (ISRO PS: SIH26168, Smart Vehicles); **4th Place**, Hacktoberfest 2026; **Research Excellence Award** (CU); Winner, ZeroToOne & Zinnovatio Hackathons.

Certs Scientific Writing (Polytechnique), Azure AI Cert (Vision, NLP).